10.05.2024

EE2004 MICROPROCESSORS MIDTERM MAKE_UP EXAM

Name: SOLUTION

CLOSED BOOK, CLOSED NOTES EXAM. NO CALCULATORS OR CHEAT SHEETS ALLOWED.
PLEASE PROVIDE YOUR ANSWERS IN DESIGNATED PLACES.
DURATION: 60 min.

Q1) (10 points) A DAC is used to output a sine wave using SysTick Interrupts and a sine-wave table. Assume the DAC has 7 bits, the DAC output is connected to a speaker, the SysTick ISR executes at 32kHz, the sine table has 256 elements, and one DAC output occurs each interrupt. The DAC output range is 0 to 3.3V. The bus clock is 80 MHz. The ADC maximum rate is 125 kHz. What frequency sound is produced, in Hz?

One output occurs at each SysTick interrupt

$\frac{32000}{256} = 125$ Hz sound is produced.

Q2) (10 points) An embedded system will use an ADC to capture electrocardiogram (EKG) data. The frequency range of the human EKG spans from 0.1 Hz to 100 Hz. What is the slowest rate at which we could sample the ADC and still have a faithful representation of the EKG in the digital samples? Give your answer as the time between samples.

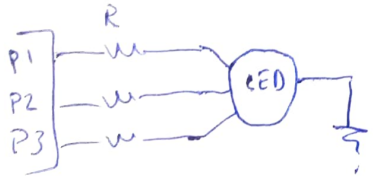
Sampling rate should be at least $2 \times f_{max}$,
that is $f_s > 200$ Hz. Slowest rate is 200 Hz.
$1/200 = 5$ ms $\therefore$ Time between samples should be less than 5 ms.

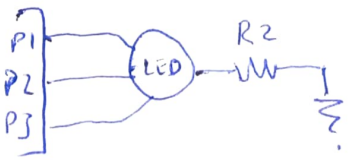
Q3) (10 points) A DAC has a range of 0 to 3V and needs a resolution of 1mV. How many bits are required? In other words, what is the smallest number of DAC bits that would satisfy the requirements?

Precision: $\frac{3-0}{0.001} = 3000$ levels needed for 1mV resolution.

11 bit gives $2^{11} = 2048$, 12 bit gives $2^{12} = 4096$ levels.
So, the DAC should operate in 12 bits resolution.

Q4) (10 points) Interface an RGB LED to the microcontroller. Operating point of each LEDs is 2V, 13mA. Each color is controlled by a separate GPIO pin with an output voltage of V = 3.3V. Specify values for resistors needed. Show equations of your calculations used to select resistor values.





$R = \frac{3.3-2}{13} = 100\,\Omega$

$R2 = \frac{3.3-2}{3 \times 13} = 33\,\Omega$

Q5) (10 points) Consider the following 8-bit addition (assume registers are 8 bits wide);

```
Load 0x40 into R1
Load 0x4B into R2
Add  R3 = R1+R2
```

a. What will be the 8-bit result in Register R3 (in hex)?

0x8B

b. What is 8-bit result in Register R3 (as unsigned decimal)?

128 + 11 = 139

c. What is 8-bit result in Register R3 (as signed decimal)?

-117

d. What will be the value of the carry (C) bit?

C = 0

e. What will be the value of the carry (V) bit?

V = 1

Q6) (10 points) Write assembly description for the following output.

Address	Value
0×20000400	0×F0F1F2F3
0×20000404	0×F4F5F6F7
0×20000408	0×F8F9FAFB
0×2000040C	0×FCFDFEFF

```
Loc1  EQU  0x20000400

      LDR  R0, =Loc1
      LDR  R1, =0XF0F1F2F3
      MOV  R2, #4
      LDR  R3, =0x04040404

Loop1 STR  R1, [R0]
      SUB  R2, R2, #1
      ADD  R0, R0, #4
      ADD  R1, R1, R3
      CMP  R2, #0
      BGT  Loop1
```

Q7) (10 points) Write assembly description of the C code given.

```
int sum = 0;
for(int i = 1; i <= 10; i++){
    sum = sum + i;
}
```

```
mov  R1, #0   ; index
mov  R0, #0   ; sum
loop add  R0, R0, R1
add  R1, R1, #1
cmp  R1, #10
ble  loop
```

Q8) (15 points) Write C code for the assembly below.

```
main
        mov     r0, #35
        mov     r1, #55
        adr     r5, 0x300
        bl      GCD

GCD
        cmp     r0,r1
        beq     halt
        subgt   r0,r0,r1
        sublt   r1,r1,r0
        b       GCD

halt
        str     r0, [r5]
```

```
int gcd(a,b) {
  while (a != b) {
    if (a > b) {
      a = a - b;
    } else {
      b = b - a;
    }
  }
  return a
}
int x = 35;
int y = 55;
int z;
z = gcd(x,y);
```

Q9) (10 points) Fill values of 'a' after execution of the C code.

Listing 9.6 Array usage as pointer example

```
int a[5] = {1, 2, 3, 4, 5};
int *aPointer;

int main(void)
{

aPointer = a;
aPointer += 1;
*aPointer = 0;

while (1);
}
```

Expressions

Expression	Type	Value
a	int [5]	0x20000000 <a>
(x) a[0]	int	1
(x) a[1]	int	0
(x) a[2]	int	3
(x) a[3]	int	4
(x) a[4]	int	5
a_pointer	int *	0x20000004 <a+4>
(x) *a_pointer	int	0

Q10) (10 points) Fill empty values after execution of the C code.

```
1  unsigned int a;
2  unsigned char b1,b2,b3,b4;
3
4  int *mem_loc;//pointer to the memory location
5
6  int main(void)
7  {
8  mem_loc= (int *) 0x20000040; //memory location to be reached
9
10 *mem_loc=0x87654321; //modify the content of memory location
11
12 a=*mem_loc;//acquire the content of memory location
13
14 //a>>n operation right shifts the content of variable a by n bits.
15
16 b1=a;
17 b2=a>>8;
18 b3=a>>16;
19 b4=a>>24;
20
21 mem_loc+=1;//point to the next location
22
23     for(;;);
24 }
```

Expressions | Modules | Live Expressions

Expression	Type	Value
(x)= a	unsigned i...	0x87654321 (Hex)
mem_loc	int *	0x20000044
(x)= b1	unsigned ...	0x21
(x)= b2	unsigned ...	0x43
(x)= b3	unsigned ...	0x65
(x)= b4	unsigned ...	0x87
Add new exp		